

Simple Networking Devices (4 PC , 2 Switchs , 1 Router)

1. PERSONAL COMPUTER (PC)

1. A **PC** (end device) is used by the user to send and receive data over the network.
 2. It works on **Layer-7 (Application Layer)** but includes all layers of the **OSI Model** for communication.
 3. Has a **Network Interface Card (NIC)** to connect physically to a network (wired or wireless).
 4. Assigned an **IP address, subnet mask, and gateway** for network identification and communication.
 5. Communicates with other devices using protocols like **HTTP, FTP, DNS, ICMP (Ping)**, etc.
 6. Can act as a **client or server** depending on the network configuration.
 7. Uses **operating system network utilities** (e.g., **ping, tracert, ipconfig, ifconfig**) for testing and troubleshooting.
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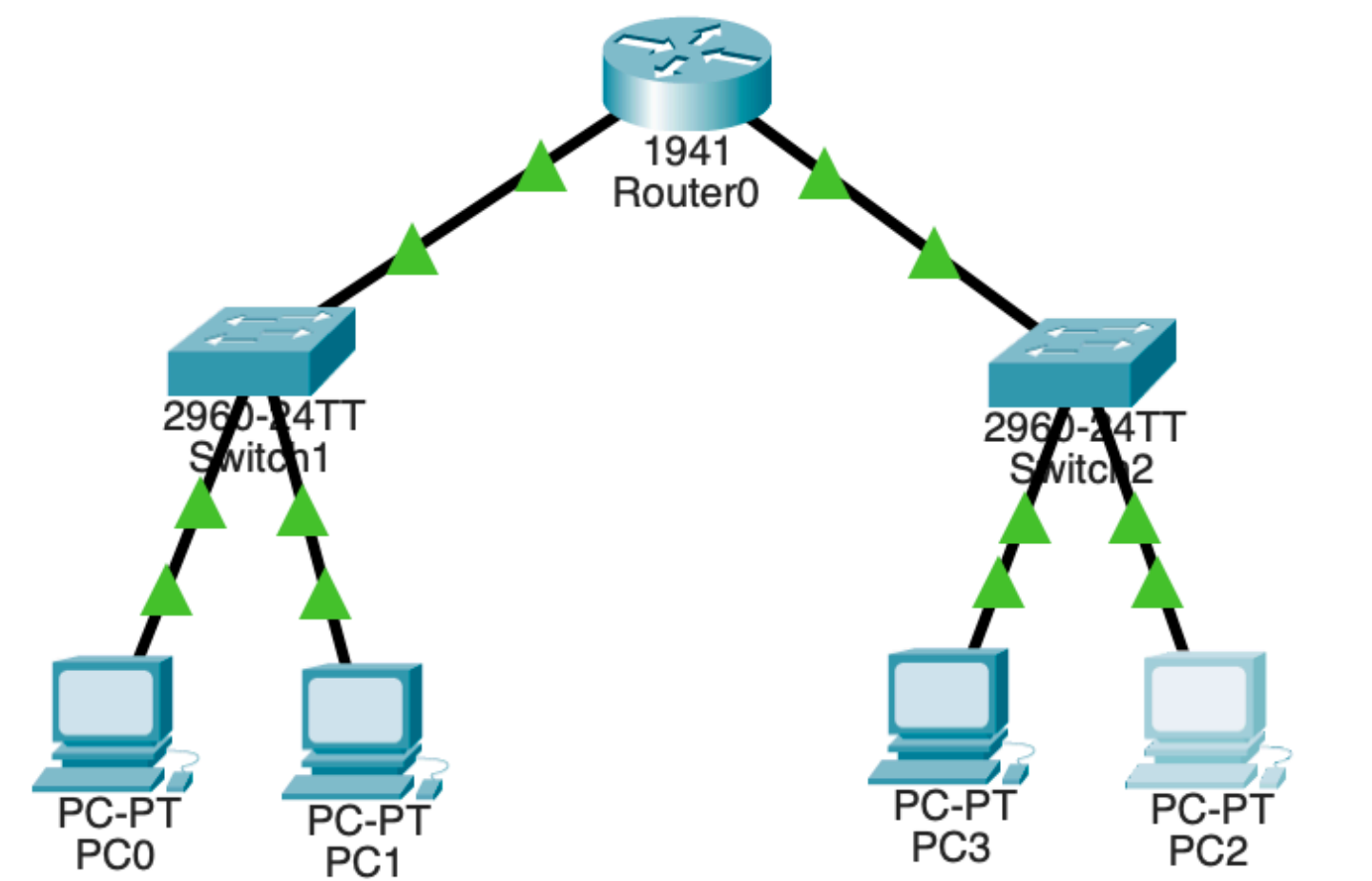
2. SWITCH

1. A **Switch** is a Layer-2 (Data Link Layer) device used to connect multiple devices within the **same local network (LAN)**.
 2. It uses **MAC addresses** to forward frames to the correct destination port, reducing network collisions.
 3. Operates in **Full-Duplex mode**, allowing simultaneous send and receive operations.
 4. Builds and maintains a **MAC address table** to remember which device is on which port.
 5. Supports **VLANs (Virtual LANs)** for logical segmentation of networks.
 6. Unlike a hub, a switch sends data only to the intended device, improving **network efficiency** and **bandwidth usage**.
 7. Can operate at Layer 3 (known as a **Layer-3 Switch**) to perform limited routing functions.
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1. ROUTER

1. A **Router** is a Layer-3 (Network Layer) device that connects **multiple networks** and directs data packets based on their **IP addresses**.
2. It determines the **best path** for data to travel using **routing tables** and **routing protocols** (e.g., **RIP, OSPF, EIGRP**).
3. Routers can connect **different LANs, WANs, or VLANs** together.
4. They use **logical addressing (IP address)** rather than MAC addresses for forwarding.
5. Provides **Network Address Translation (NAT)** and **Dynamic Host Configuration Protocol (DHCP)** functions in many setups.
6. Each router interface (Ethernet, Serial, etc.) belongs to a different **network segment**.
7. Can perform **packet filtering and traffic control**, improving network security.

Network Topology Overview



Devices:

- 1 Router → Router0 (Cisco 1941)
- 2 Switches → Switch1 and Switch2
- 4 PCs → PC0, PC1, PC2, PC3

1. Connections

From	To	Cable Type	Interface Used
Router0	Switch1	Straight-through cable	GigabitEthernet0/0 ↔ FastEthernet0/1
Router0	Switch2	Straight-through cable	GigabitEthernet0/1 ↔ FastEthernet0/1
PC0	Switch1	Straight-through cable	FastEthernet0 ↔ FastEthernet0/2
PC1	Switch1	Straight-through cable	FastEthernet0 ↔ FastEthernet0/3

From	To	Cable Type	Interface Used
PC2	Switch2	Straight-through cable	FastEthernet0 ↔ FastEthernet0/2
PC3	Switch2	Straight-through cable	FastEthernet0 ↔ FastEthernet0/3

2. Assign IP Addressing

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0 (LAN 1)	GigabitEthernet0/0	192.168.1.1	255.255.255.0	—
Router0 (LAN 2)	GigabitEthernet0/1	192.168.2.1	255.255.255.0	—
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

3. Router Configuration (in CLI)

in Router GigabitEthernet dont forget to on (both 0 and 1) this gives no shutdown state On **Router0**, enter the following commands:

```
Router> enable
Router# configure terminal

Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface gigabitEthernet0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# end
Router# write memory
```

4. Configure PCs

For each PC:

1. Go to **Desktop** → **IP Configuration**

2. Enter IP address, Subnet Mask, and Default Gateway as shown in the table above.

5. Test Connectivity

- From PC0, open **Command Prompt** → type:

```
ping 192.168.1.3    (to PC1)
ping 192.168.2.2    (to PC2)
ping 192.168.2.3    (to PC3)
```

Concept Summary

Component	Function
Router	Connects two networks (192.168.1.0/24 and 192.168.2.0/24)
Switches	Connect multiple PCs in each LAN
PCs	End devices used to test communication
Cables	Straight-through for all (since connecting different device types)